

INTERFERENCE OF LIGHT WAVES



Wave Optics

- Wave optics is a study concerned with phenomena that cannot be adequately explained by geometric (ray) optics
- These phenomena include:
 - Interference
 - Diffraction
 - Polarization

COHERENT SOURCES

- Two or more waves maintain a constant phase difference over a long distance and time, then they are said to be coherent
- The sources of light which emit light waves continuously of same
 - wavelength, and
 - time period,
 - frequency and
 - amplitude and
 - have zero phase difference or
 - constant phase difference are coherent sources.

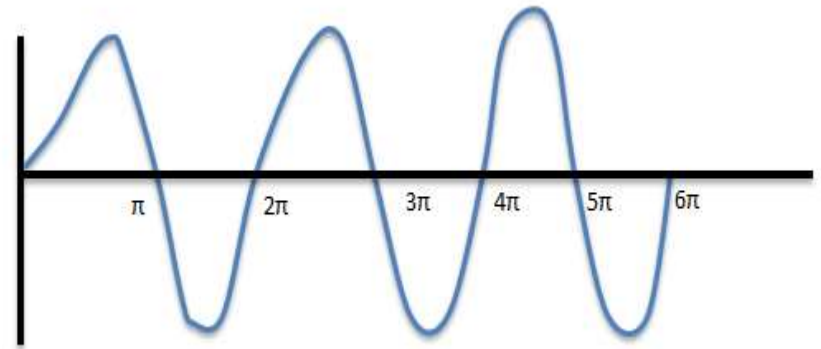
Phase difference

- Phase difference is how far 'out of sync' the two waves are
$$\Delta(\text{phase}) = \frac{2\pi\Delta(\text{path})}{\lambda}$$
- **In-Phase:** When coherent waves rise or fall together, reaching the crest (or trough) at the same time, they are said to be in-phase
- The phase difference will be 0 or 2π radians, which remains constant as the waves propagate in space
- Two waves which are in phase superimpose to give double the displacement at that point (Reinforcement / Constructive interference)

Out of Phase Points

- Out of phase points are points that are not located in the same position in the wave cycle.
- Waves that are out of phase are usually separated by an odd multiple of half wavelengths (π).
- These waves undergo destructive interference.

- π to 3π would be **inphase** because the difference is an even number 2π
- 2π to 4π would be **inphase** because the difference is also an even number 2π
- 2π to 6π would be **inphase** because the difference is an even number 4π
- 1π to 4π would be **out of phase** because it is separated by an odd number 3π .



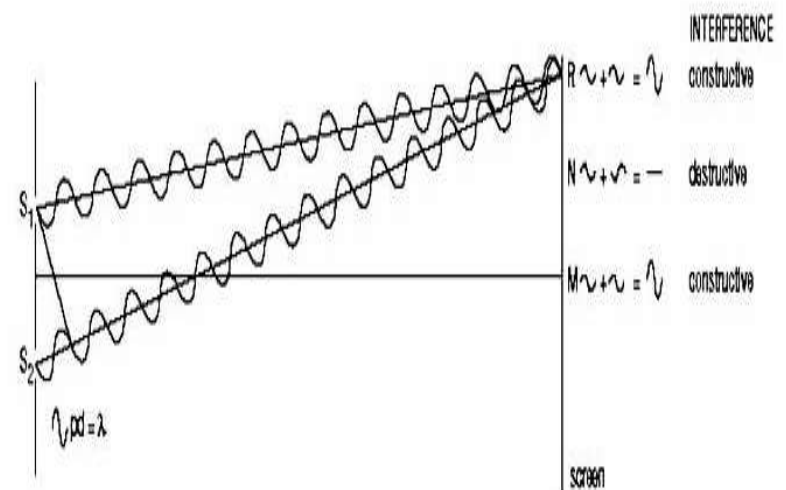
Path difference

- Path difference is the difference (in meters) between the lengths of two paths.
- When two waves have the same frequency, and travel at the same velocity, and travel the same distance they will keep the same phase difference.

- The waves emanating from S1 has arrived exactly one cycle earlier than the waves from S2
- Thus, we say that, there is a path difference between the two waves of about λ (wavelength).
- If the distance travelled by the waves from two disturbance is same, then path difference will be zero.
- Once you know the path difference, you can find the phase difference using the formula given below:

$$\Delta X = (\lambda \cdot \Delta \phi) / 2\pi$$

Here, ΔX is path difference, $\Delta \phi$ is phase difference.



Characteristics of the coherent source

The characteristics of the coherent are given below.

- Coherent source have same frequency, wavelength and amplitude
- Coherent source have same phase difference or constant phase difference.
- They can be realized in practice in:
 - a. Lloyd's single mirror (real narrow source and virtual image produced by reflection)
 - b. Fresnel bi prism (two virtual image produced by the same source by reflection)
 - c. Michelson's interferometer (dividing amplitude portion of the wave front into two parts by reflection and refraction)

Two divisions of Coherent Source

- **Division of Wave front :-**
 - Incident **wave front** is divided in to two parts by phenomenon of reflection or refraction. e.g. Lloyd's Mirror, Fresnel bi-prism, Young's double slits, Fresnel's double mirror, etc.
- **Division of Amplitude:-**
 - Incident light **amplitude** is divided in to two parts by phenomenon of reflection or refraction. Examples: Optical elements such as beam splitters and mirrors, Interference in thin film like wedge shape, Newton's ring, Michelson's interferometer

Wavefront splitting

- In this method light wave front emerging from a narrow slit pass through two closed slits.
- The two parts of the same wavefront travel through different paths and reunite on a screen to produce fringe pattern.
- This is known as interference due to division of wavefront.
- This method is useful only with narrow sources.

Amplitude splitting

- The amplitude (intensity) of a light wave is divided into two parts as reflected and transmitted components, by partial reflection at a surface.
- The two parts travel through different path and reunite to produced interference fringes.
- It is known as a interference due to division of amplitude.

Principle of superposition of waves

- When two or more progressive wave travels simultaneously in a medium without affecting the motion of each other,
- the resultant displacement of each particle of the medium at any instant of time is equal to the vector sum of the displacement produced by the two waves separately.
- This is the principle of superposition of light

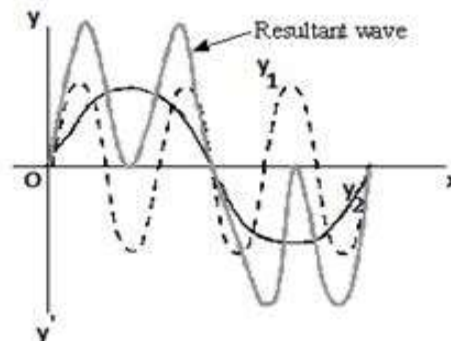


Figure : Superpositon of waves

Principle of superposition of waves

- The principle of superposition of waves can be explained as:
- a. When two waves having same frequency travel in same direction meet each other, constructive interference is formed.
- b. If there are number of waves, they travel independently and characteristics of the wave remain unchanged.
- c. The resultant intensity at any point in the medium is not equal to sum of intensity of the two waves. In constructive interference, the total intensity is greater than sum of intensities of individual waves whereas in destructive interference the total intensities is less than the sum of individual intensities.
- d. Energy distribution follows the law of conservation of energy.

Interference

- In Physics, **interference** is a phenomenon in which two waves superpose to form a resultant wave of greater, lower, or the same amplitude
- Interference usually refers to the interaction of waves that are correlated or coherent with each other, either because they come from the same source or because they have the same or nearly the same frequency
- Interference effects can be observed with all types of waves, for example, light, radio, acoustic, surface water waves or matter waves

Interference

- If two or more light waves of the same frequency overlap at a point, the resultant effect depends on the phase of the waves as well as their amplitudes
- The resultant wave at any point at any instant of time is governed by the principle of superposition
- The combined effect at each point of the region of superposition is obtained by adding algebraically the amplitudes of the individual waves

CONDITIONS FOR INTERFERENCE

- To observe interference in light waves, the following two conditions must be met:
 - 1) The sources must be **coherent**
 - They must maintain a constant phase with respect to each other
 - 2) The sources should be **monochromatic**
 - Monochromatic means they have a single wavelength

TYPES OF INTERFERENCE

- There are two types of interference

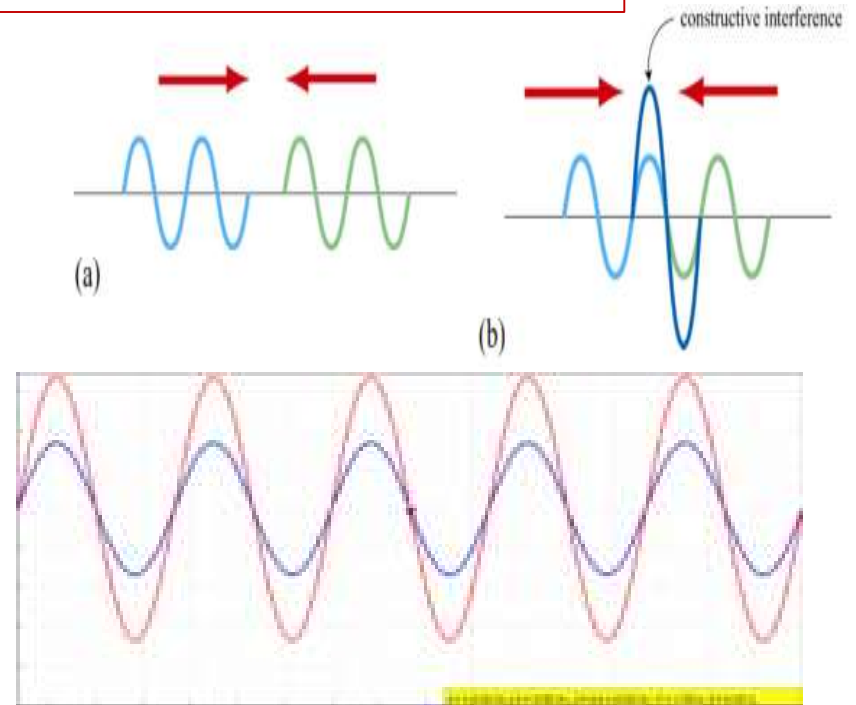
Constructive interference

Destructive interference

CONSTRUCTIVE INTERFERENCE

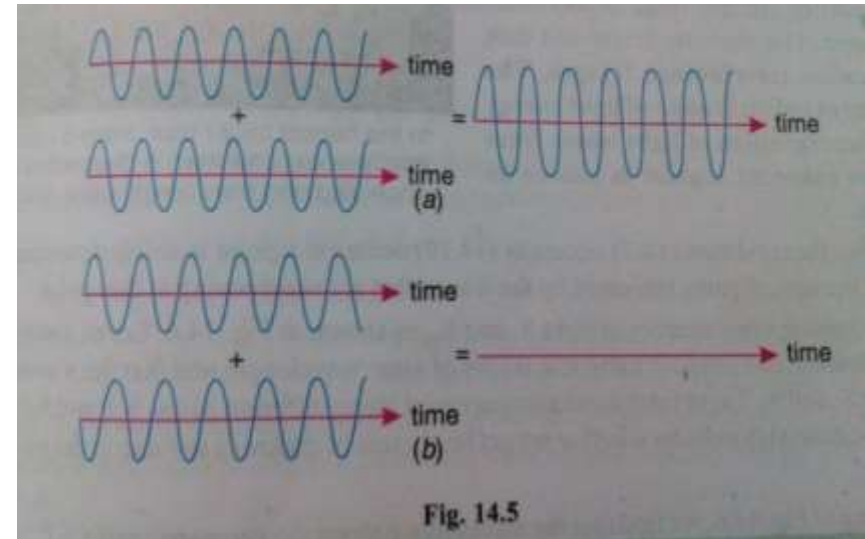
- When two light waves superpose with each other in such a way that
 - the crest of one wave falls on the crest of the second wave, and
 - trough of one wave falls on the trough of the second wave,
 - then the resultant wave has larger amplitude and it is called constructive interference.

In-phase coherence
(constructive interference)



left travelling (green) and right travelling (blue) waves in one dimension, resulting in final (red) wave

- The amplitude of the resultant wave will then be equal to the sum of the amplitudes of the two waves, as shown in Fig 14.5 (a)
- Thus the amplitude of the resultant wave
- $A_R = A + A = 2A$
- Hence, the intensity of the resultant wave is
- $I_R \propto A_R^2 = 2^2 A^2 = 2^2 I$
- It is obvious that the resultant intensity is greater than the sum of the intensities due to individual waves
- $I_R > I + I = 2I$
- Therefore the interference produced at these points is known as constructive interference

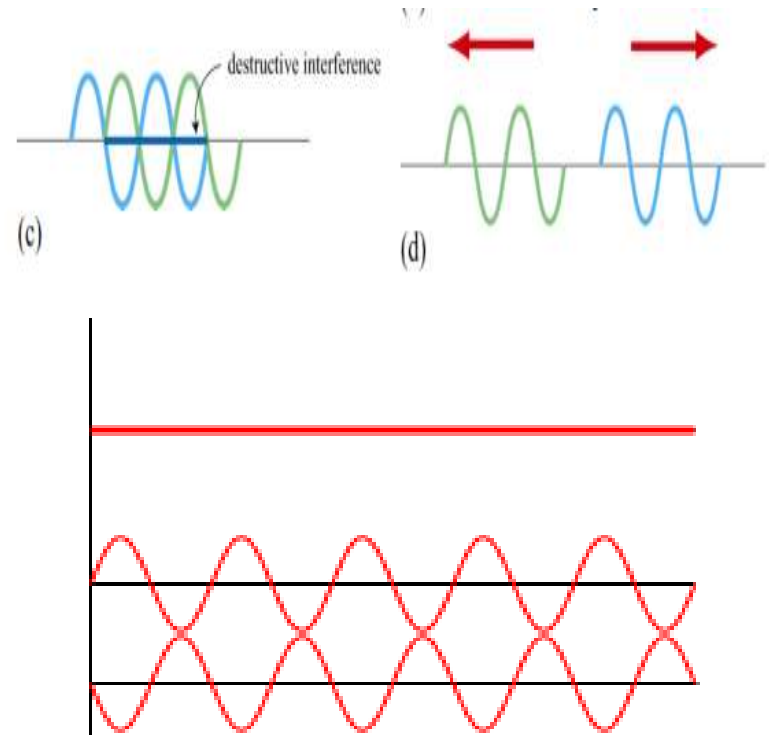


EFFECTS

- In constructive interference, two waves of light reinforce each other.
- In constructive interference, a bright fringe is obtained on the screen.
- In constructive interference the amplitude of the resultant wave is greater than that of either individual wave
- Constructive interference occurs when the phase difference between the waves is an even multiple of π (180°) (a multiple of 2π , 360°)

DESTRUCTIVE INTERFERENCE

- When two light waves superpose with each other in such a way that
 - the crest of one wave coincides the trough of the second wave,
 - then the amplitude of resultant wave becomes zero and it is called destructive interference.



Anti-phase coherence
(destructive interference)

- At certain points, the two waves may be in opposite phase
- The amplitude of the resultant will then be equal to the sum of the amplitudes of the two waves as shown in Fig 14.1 (b)
- Thus the amplitude of the resultant wave is $A_R = A - A = 0$
- Hence the intensity of the resultant wave is $I_R \propto 0$
- $I_R < 2I$

EFFECTS

- In destructive interference, two waves cancel the effects of each other
- Due to destructive interference a dark fringe is obtained on the screen.
- In *destructive interference the amplitude of the* resultant wave is less than that of either individual wave
- destructive interference occurs when the difference is an **odd multiple** of π

Young's double slit experiment



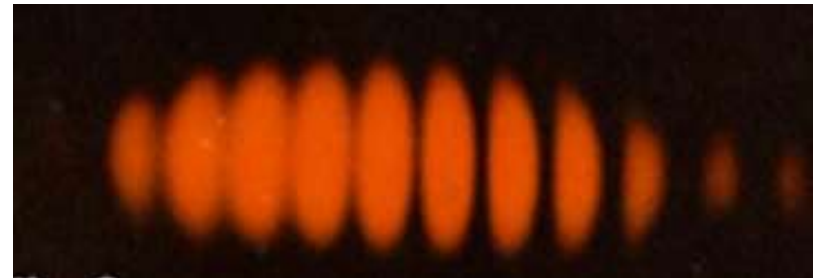
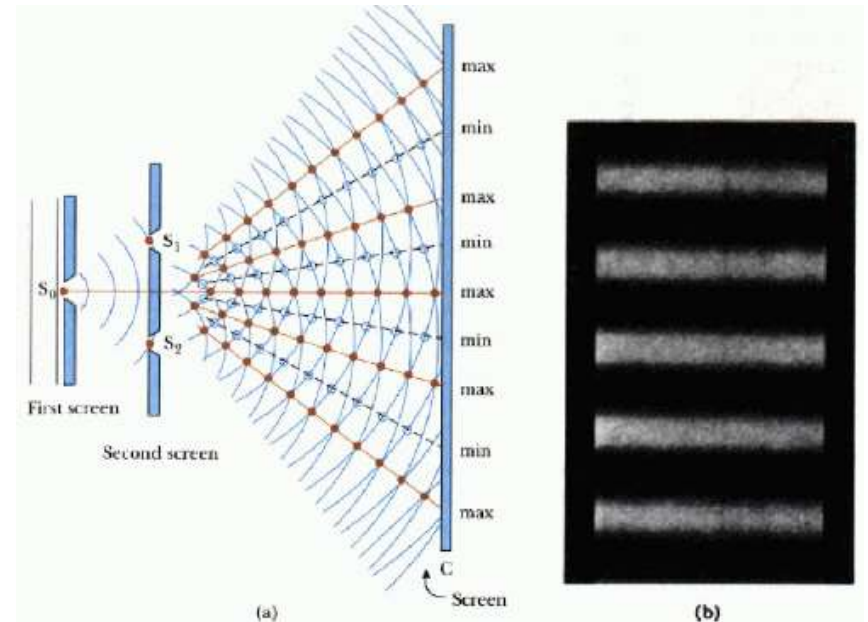
Thomas Young
(1773 – 1829) English physicist, physician and Egyptologist. Young worked on a wide variety of scientific problems, ranging from the structure of the eye and the mechanism of vision to the decipherment of the Rosetta stone. He revived the wave theory of light and recognised that interference phenomena provide proof of the wave properties of light.

Young's double slit experiment

- Thomas Young gave the first demonstration of the interference of light waves
- Young admitted the sunlight through a single pinhole and then directed the emerging light onto the pinholes
- Finally the light waves received on a screen
- The spherical waves emerging from the pinholes interfered with each other and a few colored fringes were observed on the screen

- The amount of light that emerged from the pinhole was very small and the fringes were faint and difficult to observe
- The pinholes were later replaced with narrow slits that let through much more light
- The sunlight was replaced by monochromatic light
- Young's experiment is known as double slit experiment

- Figure shows a plane view of the basic arrangement of the double slit experiment
- The primary light source is a monochromatic source
- It is generally, a sodium lamp, which emits yellow light of wavelength at around $5893 \text{ \AA}$
- This light is not suitable for causing interference because emissions from different parts of any ordinary source are not coherent



- Therefore, the monochromatic light is allowed to pass through a narrow slit at S.
- The light coming out of the slit originated only from a small region of the light source and hence behaves more nearly like an ideal light source.
- Cylindrical wavefronts are produced from the slit S, the primary light source, which fall on the two narrow closely spaced slits, S1, and S2
- The slits at S1, and S2 are very narrow.

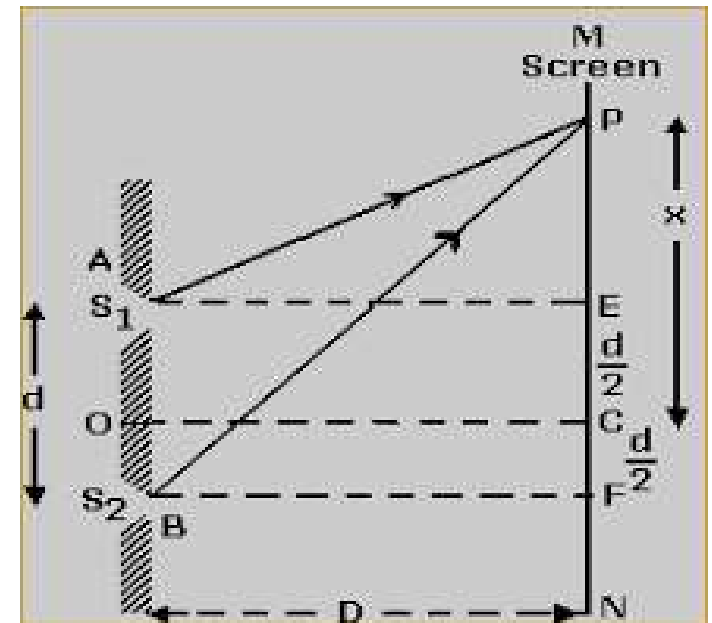
- The cylindrical waves emerging from the slits overlap.
- If the slits are equidistant from S, the phase of the wave at S1, will be the same as the phase of the wave at S2
- Further, waves leaving S1, and S2 are therefore always in phase.
- Hence, sources S1, and S2 act as secondary coherent sources.
- The waves leaving from S1 and S2 interfere and produce alternate bright and dark bands on the screen at T.

Optical path difference between the waves at P

$$(S_2P)^2 - (S_1P)^2 = D^2 + x + \frac{d^2}{2} - D^2 + x - \frac{d^2}{2} = 2xd$$

where $S_1S_2 = d$ and $OP = x$. Thus

$$S_2P - S_1P = \frac{2xd}{S_2P + S_1P}$$



$S_2P + S_1P$ (in the denominator) is replaced by $2D$.

Thus, the path difference $\delta = xd/D$

Bright Fringes

- By the principle of interference, condition for constructive interference is the path difference = $n\lambda$
- $x d / D = n\lambda$
- Here, $n = 0, 1, 2, \dots$ indicate the order of bright fringes
- So, $x = (D/d) n\lambda$
- This equation gives the distance of the n^{th} bright fringe from the point O.

Dark Fringes

- By the principle of interference, condition for destructive interference is the path difference = $(2n-1)\lambda/2$
- Here, $n = 1, 2, 3 \dots$ indicate the order of the dark fringes.
- So, $x = (D/d) [(2n - 1)\lambda/2]$
- This equation gives the distance of the n^{th} dark fringe from the point O.
- Thus, on the screen alternate dark and bright bands are seen on either side of the central bright band.

Band Width (β)

- The distance between any two consecutive bright or dark bands is called bandwidth or fringe width and is the same everywhere on the screen
- The distance between $(n+1)^{\text{th}}$ and n^{th} order consecutive bright fringes from O is given by,
- $x_{n+1} - x_n = [(D/d) [(n+1)\lambda] - (D/d) [(n)\lambda]] = (D/d) \lambda$
- Bandwidth, $\beta = (D/d) \lambda$
- Similarly, it can be proved that the distance between two consecutive dark bands is also equal to $(D/d) \lambda$.
- Since bright and dark fringes are of same width, they are equi-spaced on either side of central maximum.

Condition for Obtaining Clear and Broad Interference Bands

- The screen should be as far away from the source as possible.
- The wavelength of light used must be larger.
- The two coherent sources must be as close as possible.

Question 1

- In Young's experiment be performed with two slits in water instead of in air:
- (a) The fringes will be smaller in number
- (b) The fringes will be broader
- (c) The fringes will be narrower
- (d) No fringes will be obtained

- (a) The fringes will be smaller in number

Question 2

- In Young's interference experiment with one source and two slits, one slit is covered with a cello phone sheet so that half the intensity is absorbed. Then:
 - (a) No fringe is obtained
 - (b) Bright fringe will be brighter and dark fringe is darker
 - (c) All fringes will be darker
 - (d) Bright fringes will be less bright and dark fringes will be less dark.

- (c) All fringes will be darker

Question 3

- In Young's interference experiment, the separation between the slits is halved and the distance between slits and the screen is doubled. The fringe width is:
 - (a) Unchanged (b) Halved
 - (c) Doubled (d) Quadrupled

- (d) Quadrupled

Question 4

- In Young's experiment for interference of light with two slits, reinforcement takes place when $\sin \theta = m\lambda/d$, d is,
- (a) distance from slit to screen
- (b) distance between dark and bright fringe
- (c) distance between the slits
- (d) width of m^{th} fringe.

- (c) distance between the slits

Question 5

- In Young's double slit experiment, the distance between the slits is gradually increased. The width of the fringes system:
 - (a) increases (b) decreases
 - (c) remains the same (d) halved

- (a) increases

Thank you